

CBSE Class 12 Maths — Vectors and Three-Dimensional Geometry

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

- Q1.** Find the projection of the vector $\hat{i} - \hat{j}$ on the vector $\hat{i} + \hat{j}$. [PYQ 2023] [Confidence: Very High]
- Q2.** Write the direction cosines of a line that makes equal angles with the three coordinate axes. [PYQ 2020] [Confidence: Very High]
- Q3.** Find the value of λ for which the vectors $2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + 3\hat{k}$ are orthogonal. [PYQ 2019] [Confidence: High]
- Q4.** Find $|\vec{a} \times \vec{b}|$, if $|\vec{a}| = 2$, $|\vec{b}| = 7$, and $\vec{a} \cdot \vec{b} = 7$. [PYQ 2021] [Confidence: High]
- Q5.** Write the Cartesian equation of the z-axis in 3D space. [Practice] [Confidence: Medium]
- Q6.** Find the unit vector in the direction of the vector $\vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}$. [PYQ 2022] [Confidence: Very High]
- Q7.** Find the perpendicular distance of the point $P(a, b, c)$ from the x-axis. [Practice] [Confidence: Medium]
- Q8.** If $\vec{a}$ and $\vec{b}$ are unit vectors and the angle between them is θ , what is the value of $|\vec{a} - \vec{b}|$ in terms of θ ? [PYQ 2018] [Confidence: High]
- Q9.** Determine the direction cosines of the line passing through the origin and the point $(1, -2, 3)$. [PYQ 2020] [Confidence: High]
- Q10.** What is the value of $\hat{i} \cdot (\hat{j} \times \hat{k}) + \hat{j} \cdot (\hat{i} \times \hat{k}) + \hat{k} \cdot (\hat{i} \times \hat{j})$? [PYQ 2016] [Confidence: Very High]
- Q11.** Find the area of the parallelogram whose adjacent sides are determined by the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $2\hat{i} + \hat{j} - 4\hat{k}$. [PYQ 2019] [Confidence: High]
- Q12.** Are the points with position vectors $\vec{a} - 2\vec{b} + 3\vec{c}$, $-2\vec{a} + 3\vec{b} + 2\vec{c}$ and $-8\vec{a} + 13\vec{b}$ collinear? (Yes/No) [Practice] [Confidence: Medium]
- Q13.** Write the vector equation of the straight line given by Cartesian equations $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$. [PYQ 2017] [Confidence: Very High]
- Q14.** For any non-zero vector $\vec{a}$, write the simplified value of $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$. [PYQ 2020] [Confidence: High]
- Q15.** Find the position vector of the midpoint of the line segment joining points $A(2, 3, 4)$ and $B(4, 1, -2)$. [PYQ 2023] [Confidence: Very High]
- Q16.** The direction ratios of a line are $2, -1, -2$. What are its direction cosines? [PYQ 2018] [Confidence: Very High]
- Q17.** Find a vector of magnitude 5 units in the direction of the resultant of vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$. [PYQ 2022] [Confidence: High]
- Q18.** If the scalar projection of vector $x\hat{i} - \hat{j} + \hat{k}$ on the vector $2\hat{i} - \hat{j} + 5\hat{k}$ is $\frac{1}{\sqrt{30}}$, find the value of x . [PYQ 2019] [Confidence: Medium]
- Q19. Assertion (A):** The vectors $\vec{p} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{q} = 3\hat{i} + 2\hat{j} - 2\hat{k}$ are orthogonal. **Reason (R):** Two non-zero vectors $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$. Choose the correct option (A, B, C, D). [Practice] [Confidence: Very High]
- Q20. Assertion (A):** The shortest distance between two intersecting lines is strictly zero. **Reason (R):** Intersecting lines share a common point, hence the minimum distance between any points on them must be zero. Choose the correct option (A, B, C, D). [PYQ 2024] [Confidence: Very High]

2-Mark

- Q21.** Find the angle between the vectors $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. [PYQ 2019] [Confidence: Very High]
- Q22.** If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, prove that the vectors $\vec{a}$ and $\vec{b}$ are perpendicular to each other. [PYQ 2020] [Confidence: Very High]
- Q23.** Find the coordinates of the point where the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+1}{4}$ intersects the xy-plane. [PYQ 2018] [Confidence: High]
- Q24.** Find the value of p for which the straight lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{p}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ are perpendicular to each other. [PYQ 2017] [Confidence: High]
- Q25.** Find a unit vector perpendicular to both the vectors $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$. [PYQ 2021] [Confidence: Very High]
- Q26.** Show that the points $A(2, 3, -4)$, $B(1, -2, 3)$ and $C(3, 8, -11)$ are collinear using vector or 3D geometry methods. [PYQ 2022] [Confidence: High]
- Q27.** Write the Cartesian equations of a line passing through the point $(1, 2, -4)$ and parallel to the vector $\hat{i} + 2\hat{j} - \hat{k}$. [PYQ 2019] [Confidence: Very High]
- Q28.** If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular vectors of the same magnitude, prove that the vector $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to the vectors $\vec{a}, \vec{b}$ and $\vec{c}$. [PYQ 2020] [Confidence: Medium]
- Q29.** Find the scalar and vector components of the vector with initial point $P(2, 1, 3)$ and terminal point $Q(-5, 7, 1)$. [Practice] [Confidence: High]
- Q30.** If a line makes angles 90° , 135° , and 45° with the positive directions of x, y, and z axes respectively, find its direction cosines. [PYQ 2023] [Confidence: Very High]
- Q31.** Find the area of a triangle with vertices $A(1, 1, 1)$, $B(1, 2, 3)$, and $C(2, 3, 1)$ using vector methods. [PYQ 2022] [Confidence: High]
- Q32.** If $|\vec{a}| = 3$, $|\vec{b}| = 4$, and $|\vec{a} \times \vec{b}| = 6$, find the angle between vectors $\vec{a}$ and $\vec{b}$. [PYQ 2019] [Confidence: High]

3-Mark

- Q33.** Find the shortest distance between the lines given by vector equations $\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$. [PYQ 2023] [Confidence: Very High]
- Q34.** Find the Cartesian equation of the line passing through the point $(-1, 3, -2)$ and perpendicular to the lines $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$. [PYQ 2020] [Confidence: High]
- Q35.** Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect each other. Also, find their point of intersection. [PYQ 2018] [Confidence: High]
- Q36.** Find the foot of the perpendicular drawn from the point $P(0, 2, 3)$ onto the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. [PYQ 2022] [Confidence: Very High]
- Q37.** Using vectors, find the exact area of the triangle with vertices $A(1, 1, 2)$, $B(2, 3, 5)$, and $C(1, 5, 5)$. [PYQ 2019] [Confidence: High]
- Q38.** If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 5$, find the value of the expression $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$. [PYQ 2021] [Confidence: Very High]
- Q39.** Find the shortest distance between the lines represented by $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$. [PYQ 2019] [Confidence: Very High]
- Q40.** Find the angle between the lines whose direction cosines are given by the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$. [PYQ 2017] [Confidence: Medium]
- Q41.** Find the value of λ so that the lines $\frac{1-x}{3} = \frac{7y-14}{2\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles. [PYQ 2020] [Confidence: High]

Q42. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$, and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{d}$ which is strictly perpendicular to both $\vec{a}$ and $\vec{b}$, and satisfies $\vec{c} \cdot \vec{d} = 15$. [PYQ 2018] [Confidence: Medium]

5-Mark (Long Answer/Case-Study)

Q43. Find the shortest distance between the skew lines whose vector equations are

$$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}) \text{ and } \vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k}). \text{ [PYQ 2023] [Confidence: Very High]}$$

Q44. Find the coordinates of the foot of the perpendicular drawn from the point $P(2, -1, 5)$ to the line $\frac{x-11}{10} = \frac{y+2}{-4} = \frac{z+8}{-11}$. Also, find the exact length of the perpendicular. [PYQ 2020] [Confidence: Very High]

Q45. Find the image of the point $(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Also, write the vector equation of the line joining the given point and its image. [PYQ 2019] [Confidence: High]

Q46. Find the shortest distance between the parallel lines given by $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(\hat{i} - \hat{j} + \hat{k})$. [PYQ 2018] [Confidence: Very High]

Q47. Case Study (Vector Algebra): A farmer uses three agricultural drones to spray fertilizer on his triangular field. The drones are stationed at three points in the field representing the vertices of the triangle, defined by the coordinates $A(1, 1, 1)$, $B(2, 3, 5)$, and $C(1, 5, 5)$. (i) Find the position vector representing the displacement $\vec{AB}$ and its magnitude. (1 Mark) (ii) Find the cross product $\vec{AB} \times \vec{AC}$. (2 Marks) (iii) Using vector methods, calculate the total area of the triangular field ABC covered by the drones. (2 Marks) [Practice] [Confidence: High]

Q48. Case Study (3D Geometry): Two flights are flying in the air along two straight-line paths. Flight 1 follows the path $\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$ and Flight 2 follows the path $\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} + (2s+1)\hat{k}$. (i) Write the direction ratios of the paths of Flight 1 and Flight 2. (1 Mark) (ii) Determine mathematically whether the paths of the two flights are parallel, intersecting, or skew lines. (2 Marks) (iii) Calculate the shortest distance between the two flight paths to ensure they don't collide. (2 Marks) [Practice] [Confidence: Very High]

Q49. Find the vector and Cartesian equations of the line passing through $(1, 2, -4)$ and perpendicular to the two lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$. [PYQ 2021] [Confidence: High]

Q50. Find the equation of the line which intersects the lines $\frac{x+2}{1} = \frac{y-3}{2} = \frac{z+1}{4}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and passes through the point $(1, 1, 1)$. [PYQ 2018] [Confidence: Medium]

Answer Key

Q1. 0. ($\vec{a} \cdot \vec{b} = 1(1) + (-1)(1) = 0$, so projection is 0). **Q2.** $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$. **Q3.**
 $\vec{a} \cdot \vec{b} = 2(1) + \lambda(-2) + 1(3) = 0 \implies 5 - 2\lambda = 0 \implies \lambda = \frac{5}{2}$. **Q4.** $\cos \theta = \frac{7}{14} = \frac{1}{2} \implies \theta = 60^\circ$.
 $|\vec{a} \times \vec{b}| = 2 \times 7 \times \sin 60^\circ = 7\sqrt{3}$. **Q5.** $\frac{x-0}{0} = \frac{y-0}{0} = \frac{z-0}{1}$. **Q6.** $\hat{a} = \frac{2\hat{i}+3\hat{j}+\hat{k}}{\sqrt{14}}$. **Q7.** Distance from x-axis is $\sqrt{b^2 + c^2}$. **Q8.** $|\vec{a} - \vec{b}|^2 = 1^2 + 1^2 - 2(1)(1)\cos \theta = 2(1 - \cos \theta) = 4\sin^2(\theta/2)$. So, $|\vec{a} - \vec{b}| = 2\sin(\theta/2)$. **Q9.** Direction ratios: 1, -2, 3. Direction cosines: $\frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$. **Q10.** $1 - 1 + 1 = 1$. **Q11.** Area = $|\vec{a} \times \vec{b}| = |5\hat{i} + 10\hat{j} + 5\hat{k}| = \sqrt{25 + 100 + 25} = \sqrt{150} = 5\sqrt{6}$ sq units. **Q12.** Yes. Let the points be P, Q, R. $\vec{PQ} = -3\vec{a} + 5\vec{b} - \vec{c}$. $\vec{QR} = -6\vec{a} + 10\vec{b} - 2\vec{c} = 2\vec{PQ}$. Since vectors are proportional and share point Q, they are collinear. **Q13.** $\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} + 2\hat{k})$. **Q14.** Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$. Result evaluates to $2(x^2 + y^2 + z^2) = 2|\vec{a}|^2$. **Q15.** Midpoint = $\frac{2+4}{2}\hat{i} + \frac{3+1}{2}\hat{j} + \frac{4-2}{2}\hat{k} = 3\hat{i} + 2\hat{j} + \hat{k}$. **Q16.** Magnitude = $\sqrt{4 + 1 + 4} = 3$. DCs are $\frac{2}{3}, \frac{-1}{3}, \frac{-2}{3}$. **Q17.** Resultant $\vec{R} = \vec{a} + \vec{b} = 3\hat{i} + \hat{j}$. Magnitude = $\sqrt{10}$. Required vector = $\frac{5}{\sqrt{10}}(3\hat{i} + \hat{j}) = \frac{\sqrt{10}}{2}(3\hat{i} + \hat{j})$. **Q18.** Projection = $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{2x+1+5}{\sqrt{4+1+25}} = \frac{2x+6}{\sqrt{30}}$. So $2x + 6 = 1 \implies x = -5/2$. **Q19.** Option A. Both are true and R is the correct explanation of A. (Dot product is $6 - 2 - 4 = 0$). **Q20.** Option A. Both are true and R is the correct explanation of A.

Q21. $\cos \theta = \frac{1-1-1}{\sqrt{3}\sqrt{3}} = \frac{-1}{3} \Rightarrow \theta = \cos^{-1}(-1/3)$. **Q22.** Squaring both sides:

$|\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} \Rightarrow 4\vec{a} \cdot \vec{b} = 0 \Rightarrow \vec{a} \cdot \vec{b} = 0$, so $\vec{a} \perp \vec{b}$. **Q23.** In xy-plane, $z = 0$. So $\frac{0+1}{4} = \frac{1}{4}$. Thus $x = 2(1/4) + 1 = 3/2$ and $y = -3(1/4) + 2 = 5/4$. Point is $(3/2, 5/4, 0)$. **Q24.**

$a_1a_2 + b_1b_2 + c_1c_2 = 0 \Rightarrow 2(3) + 3(4) + p(5) = 0 \Rightarrow 18 + 5p = 0 \Rightarrow p = -18/5$. **Q25.**

$\vec{a} \times \vec{b} = \hat{i}(3) - \hat{j}(3) + \hat{k}(-3) = 3\hat{i} - 3\hat{j} - 3\hat{k}$. Unit vector $= \pm \frac{\hat{i}-\hat{j}-\hat{k}}{\sqrt{3}}$. **Q26.** Direction ratios of AB : $-1, -5, 7$.

Direction ratios of BC : $2, 10, -14$. Since $-1/2 = -5/10 = 7/-14$, $AB \parallel BC$. Since B is common, they are collinear. **Q27.** $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+4}{-1}$. **Q28.** Let $|\vec{a}| = |\vec{b}| = |\vec{c}| = k$. Let $\vec{R} = \vec{a} + \vec{b} + \vec{c}$.

$|\vec{R}|^2 = 3k^2 \Rightarrow |\vec{R}| = k\sqrt{3}$. $\cos \alpha = \frac{\vec{R} \cdot \vec{a}}{|\vec{R}||\vec{a}|} = \frac{k^2}{k\sqrt{3} \cdot k} = \frac{1}{\sqrt{3}}$. Same for β and γ . **Q29.** Vector =

$\vec{OQ} - \vec{OP} = -7\hat{i} + 6\hat{j} - 2\hat{k}$. Scalar components: $-7, 6, -2$. Vector components: $-7\hat{i}, 6\hat{j}, -2\hat{k}$. **Q30.**

$l = \cos 90^\circ = 0$, $m = \cos 135^\circ = -1/\sqrt{2}$, $n = \cos 45^\circ = 1/\sqrt{2}$. DCs are $0, -1/\sqrt{2}, 1/\sqrt{2}$. **Q31.**

$\vec{AB} = 0\hat{i} + 1\hat{j} + 2\hat{k}$, $\vec{AC} = 1\hat{i} + 2\hat{j} + 0\hat{k}$. Area $= \frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} |-4\hat{i} + 2\hat{j} - \hat{k}| = \frac{\sqrt{21}}{2}$ sq units. **Q32.**

$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin \theta \Rightarrow 6 = 12\sin \theta \Rightarrow \sin \theta = 1/2 \Rightarrow \theta = 30^\circ$ or $\pi/6$.

Q33. $\vec{a}_2 - \vec{a}_1 = -3\hat{i} + 5\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = 2\hat{i} - \hat{j} + 0\hat{k}$. $SD = \frac{|(-3\hat{i}+5\hat{k}) \cdot (2\hat{i}-\hat{j})|}{\sqrt{4+1+0}} = \frac{|-6|}{\sqrt{5}} = \frac{6}{\sqrt{5}}$ units. **Q34.** Direction

vector is parallel to $(1\hat{i} + 2\hat{j} + 3\hat{k}) \times (-3\hat{i} + 2\hat{j} + 5\hat{k}) = 4\hat{i} - 14\hat{j} + 8\hat{k}$. Simplifying DRs to $2, -7, 4$. Equation: $\frac{x+1}{2} = \frac{y-3}{-7} = \frac{z+2}{4}$. **Q35.** General points: $(3\lambda - 1, 5\lambda - 3, 7\lambda - 5)$ and $(\mu + 2, 3\mu + 4, 5\mu + 6)$. Equating and

solving gives $\lambda = 1/2, \mu = -3/2$. Both satisfy the z-coordinate. Point of intersection is $(1/2, -1/2, -3/2)$. **Q36.**

Let foot be $(5\lambda - 3, 2\lambda + 1, 3\lambda - 4)$. DRs of perpendicular: $5\lambda - 3, 2\lambda - 1, 3\lambda - 7$. Dot product with line's DRs $(5, 2, 3)$ equals 0. $25\lambda - 15 + 4\lambda - 2 + 9\lambda - 21 = 0 \Rightarrow 38\lambda = 38 \Rightarrow \lambda = 1$. Foot is $(2, 3, -1)$. **Q37.**

$\vec{AB} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{AC} = 0\hat{i} + 4\hat{j} + 3\hat{k}$. Cross product $= -6\hat{i} - 3\hat{j} + 4\hat{k}$. Magnitude $= \sqrt{36 + 9 + 16} = \sqrt{61}$.

Area $= \frac{\sqrt{61}}{2}$ sq units. **Q38.**

$(\vec{a} + \vec{b} + \vec{c})^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0 \Rightarrow 9 + 16 + 25 + 2(\Sigma) = 0 \Rightarrow \Sigma = -25$

. **Q39.** $\vec{a}_2 - \vec{a}_1 = \hat{i} + 2\hat{j} + 2\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = -\hat{i} + 2\hat{j} - \hat{k}$. $SD = \frac{|(-1+4-2)|}{\sqrt{1+4+1}} = \frac{1}{\sqrt{6}}$ units. **Q40.** $l = -(m + n)$.

Substitute into $l^2 + m^2 - n^2 = 0 \Rightarrow 2m(m + n) = 0$. Case 1: $m = 0, l = -n$, DRs $(-1, 0, 1)$. Case 2:

$m = -n, l = 0$, DRs $(0, -1, 1)$. $\cos \theta = \frac{0+0+1}{\sqrt{2}\sqrt{2}} = 1/2 \Rightarrow \theta = 60^\circ$. **Q41.** Standard forms:

$\frac{x-1}{-3} = \frac{y-2}{2\lambda/7} = \frac{z-3}{2}$ and $\frac{x-1}{-3\lambda/7} = \frac{y-5}{1} = \frac{z-6}{-5}$. Dot product of DRs =

$0 \Rightarrow 9\lambda/7 + 2\lambda/7 - 10 = 0 \Rightarrow 11\lambda/7 = 10 \Rightarrow \lambda = 70/11$. **Q42.** $\vec{d}$ is parallel to

$\vec{a} \times \vec{b} = 32\hat{i} - \hat{j} - 14\hat{k}$. Let $\vec{d} = k(32\hat{i} - \hat{j} - 14\hat{k})$.

$\vec{c} \cdot \vec{d} = 15 \Rightarrow k(64 + 1 - 56) = 15 \Rightarrow 9k = 15 \Rightarrow k = 5/3$. $\vec{d} = \frac{5}{3}(32\hat{i} - \hat{j} - 14\hat{k})$.

Q43. $\vec{a}_2 - \vec{a}_1 = \hat{i} - 3\hat{j} - 2\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = -3\hat{i} + 0\hat{j} + 3\hat{k}$. $SD = \frac{|(-3+0-6)|}{\sqrt{9+0+9}} = \frac{9}{\sqrt{18}} = \frac{3}{\sqrt{2}}$ units. **Q44.** Let foot be

$(10\lambda + 11, -4\lambda - 2, -11\lambda - 8)$. DRs of perpendicular are $10\lambda + 9, -4\lambda - 1, -11\lambda - 13$. Dot product with $(10, -4, -11) = 0 \Rightarrow 100\lambda + 90 + 16\lambda + 4 + 121\lambda + 143 = 0 \Rightarrow 237\lambda = -237 \Rightarrow \lambda = -1$. Foot

is $(1, 2, 3)$. Length $= \sqrt{1^2 + (-3)^2 + 2^2} = \sqrt{14}$ units. **Q45.** Let foot of perpendicular be $F(\lambda, 2\lambda + 1, 3\lambda + 2)$.

DRs of line to point: $\lambda - 1, 2\lambda - 5, 3\lambda - 1$. Dot product with $(1, 2, 3)$ is zero

$\Rightarrow \lambda - 1 + 4\lambda - 10 + 9\lambda - 3 = 0 \Rightarrow 14\lambda = 14 \Rightarrow \lambda = 1$. Foot is $(1, 3, 5)$. Image

$I(x, y, z) \Rightarrow (x+1)/2 = 1, (y+6)/2 = 3, (z+3)/2 = 5 \Rightarrow I(1, 0, 7)$. Line equation:

$\vec{r} = (\hat{i} + 6\hat{j} + 3\hat{k}) + \gamma(0\hat{i} - 6\hat{j} + 4\hat{k})$. **Q46.** Since $\vec{b}_1 = \vec{b}_2 = \hat{i} - \hat{j} + \hat{k}$, lines are parallel. Distance $= \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$.

$\vec{a}_2 - \vec{a}_1 = \hat{i} - 3\hat{j} - 4\hat{k}$. Cross product $= -7\hat{i} - 5\hat{j} + 2\hat{k}$. $SD = \frac{\sqrt{49+25+4}}{\sqrt{3}} = \frac{\sqrt{78}}{\sqrt{3}} = \sqrt{26}$ units. **Q47.** (i)

$\vec{AB} = \hat{i} + 2\hat{j} + 4\hat{k}$. Magnitude $= \sqrt{21}$. (ii) $\vec{AC} = 0\hat{i} + 4\hat{j} + 4\hat{k}$. $\vec{AB} \times \vec{AC} = -8\hat{i} - 4\hat{j} + 4\hat{k}$. (iii) Area =

$\frac{1}{2} |-8\hat{i} - 4\hat{j} + 4\hat{k}| = \frac{1}{2} \sqrt{64 + 16 + 16} = \frac{\sqrt{96}}{2} = 2\sqrt{6}$ sq units. **Q48.** (i) Path 1:

$\vec{r} = (\hat{i} - 2\hat{j} + 3\hat{k}) + t(-\hat{i} + \hat{j} - 2\hat{k})$. DRs: $-1, 1, -2$. Path 2: $\vec{r} = (\hat{i} - \hat{j} + \hat{k}) + s(\hat{i} + 2\hat{j} + 2\hat{k})$. DRs: $1, 2, 2$. (ii)

Vectors are not scalar multiples, so not parallel. Find SD: $\vec{a}_2 - \vec{a}_1 = \hat{j} - 2\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = 6\hat{i} + 0\hat{j} - 3\hat{k}$.

$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 6 \neq 0$. Thus, lines are skew. (iii) $SD = \frac{|6|}{\sqrt{36+9}} = \frac{6}{\sqrt{45}} = \frac{2}{\sqrt{5}}$ units. **Q49.** Normal to

required line is parallel to $(3\hat{i} - 16\hat{j} + 7\hat{k}) \times (3\hat{i} + 8\hat{j} - 5\hat{k}) = 24\hat{i} + 36\hat{j} + 72\hat{k}$. Dividing by 12, DRs are $2, 3, 6$.

Cartesian equation: $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$. Vector eq: $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$. **Q50.** Let the required

line intersect the two lines at $P(\lambda - 2, 2\lambda + 3, 4\lambda - 1)$ and $Q(2\mu + 1, 3\mu + 2, 4\mu + 3)$. Since $(1, 1, 1)$ lies on PQ, points $(1, 1, 1), P, Q$ are collinear. DRs of line $1 \rightarrow P: \lambda - 3, 2\lambda + 2, 4\lambda - 2$. DRs of line $1 \rightarrow Q: 2\mu, 3\mu + 1, 4\mu + 2$. Proportionality: $\frac{\lambda-3}{2\mu} = \frac{2\lambda+2}{3\mu+1} = \frac{4\lambda-2}{4\mu+2}$. Solving gives $\lambda = 1, \mu = 0$. Point P is $(-1, 5, 3)$. Line through $(1, 1, 1)$ and $(-1, 5, 3)$ has DRs $-2, 4, 2 \equiv -1, 2, 1$. Equation: $\frac{x-1}{-1} = \frac{y-1}{2} = \frac{z-1}{1}$.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).